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Efficient differential expression analysis of large-scale single-cell transcriptomics data using Dreamlet

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Advances in single-cell and -nucleus transcriptomics have enabled generation of increasingly large-scale datasets from hundreds of subjects and millions of cells. These studies promise to give unprecedented insight into the cell type specific biology of human disease. Yet performing differential expression analyses across subjects remains difficult due to challenges in statistical modeling of these complex studies and scaling analyses to large datasets. Our open-source R package dreamlet (DiseaseNeurogenomics.github.io/dreamlet) uses a pseudobulk approach based on precision-weighted linear mixed models to identify genes differentially expressed with traits across subjects for each cell cluster. Designed for data from large cohorts, dreamlet is substantially faster and uses less memory than existing workflows, while supporting complex statistical models and controlling the false positive rate. We demonstrate computational and statistical performance on published datasets, and a novel dataset of 1.4 M single nuclei from postmortem brains of 150 Alzheimer's disease cases and 149 controls.

The human body is composed of hundreds of cell types, each with its own role in the biology of health and disease[1]. The cell-type specificity of fundamental biological mechanisms has long been appreciated[1,2], and genome-wide association studies have revealed strong cell type-specific enrichments of risk variants that inform our understanding of disease biology[3–5].

Recent advances in single-cell-nucleus transcriptomics technology have enabled profiling of over a million cells from hundreds of subjects to study variation in gene expression associated with disease or other traits[6–11]. Multiplexing samples and assigning cells to subjects following sequencing, using genetic variation[12,13] or hashing with barcoded antibodies[14], has enabled a further increase in the scale of these studies. As the scale of single-cell data continues to increase and studies of cross-subject variation profile more subjects and cells, analytical workflows must keep pace.

Much of the work on differential expression analysis in single-cell transcriptomics has focused on identifying expression differences between cell clusters[15–17]. With the increasing scale of single-cell datasets, recent work has examined approaches for performing differential expression analysis across subjects. Existing methods for this application were designed for small to moderate-sized datasets, and these model gene expression at either the single cell or the pseudobulk level

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for each cell type cluster[18–21]. Modeling expression at the single cell level is the most direct approach and can be performed with a range of statistical models while allowing cell-level covariates[18–21]. Yet statistical modeling of cell-level counts is challenging due to the low read depth and pervasive dropout effects[18,21], and it is essential to model the fact that multiple cells are sampled from the same subject[19,20,22]. Even then, controlling the false positive rate remains challenging[18]. Moreover, as studies assay more cells, the computational cost of fitting thousands of cell-level regression models becomes prohibitive[18].

Alternatively, pseudobulk approaches aggregate reads across cells within a cell cluster and then use methods originally developed for bulk RNA-seq[18]. This works well for moderately sized datasets, but aggregating reads across cells using existing methods in large-scale studies can be very computationally and memory-intensive. More importantly, emerging single-cell and -nucleus transcriptome datasets use complex study designs including technical or biological replicates, or use sample multiplexing that can introduce high-dimensional batch effects that existing pseudobulk methods cannot model adequately. Here, we introduce dreamlet, an open-source R package that shows superior computational and statistical performance compared to existing methods, addressing previous challenges related to statistical modeling of data from large cohorts.

Results

Dreamlet workflow for large-scale differential expression analysis

Dreamlet applies a pseudobulk approach and fits a regression model for each gene and cell cluster to test for differential expression associated with trait variation across subjects. Use of precision-weighted linear mixed models enables accounting for repeated measures study designs, including technical or biological replicates, high-dimensional batch effects due to sample multiplexing, and variation in sequencing depth and cell number[23] (Fig. 1). Dreamlet incorporates precision weights at two levels to account for uncertainty in the observed gene expression measurements. Dreamlet first initializes the precision weights using a Poisson count model of the observed data that considers the fact that observing more cells and reads from a sample increases the measurement precision of the underlying expression state[18] (Supplementary Methods). Incorporating these weights does not affect computational time. These initial weights are then used to estimate an empirical mean-variance trend that estimates a second round of weights without assuming a parametric model for the counting error[24]. Dreamlet also extends the use of empirical Bayes moderated t-statistics, which borrow information across genes to increase power and control of false positive rate[25], to the case of precision-weighted linear mixed models (Supplementary Methods). Finally, the dreamlet workflow is designed for large-scale data and is substantially faster and uses less memory than existing workflows, while supporting complex statistical models and controlling the false positive rate.

Computational performance on large-scale datasets

The dreamlet workflow uses an efficient implementation to compute pseudobulk counts and scales to larger datasets than competing methods. Dreamlet processes pseudobulk for 1000 donors across eight cell types for 4.1 million cells in 31 min using only 26 Gb of memory (Fig. 2A, B). A naive method has similar memory usage, but is substantially slower. Performance of pegasus[26] is slightly faster for less than 500 donors, but its memory footprint is substantially larger and increases rapidly with sample size. Other methods are 10-100x slower or are limited by memory constraints even for moderate sample sizes.

Dreamlet fits precision-weighted linear mixed models for each gene parallelized across multiple CPU cores and enables analysis of 12 cell types across 326 subjects in 45 CPU minutes corresponding to a wall time of 15 min in this case (Fig. 2C). This is over an order of magnitude faster than using generalized linear mixed models (GLMMs) at the single cell level, and competitive with other pseudobulk approaches using a negative binomial model that are not able to model random effects (i.e., edgeR[27], DESeq2[28]).

Statistical performance on large-scale datasets

Using a simulation pipeline and benchmarks developed by an independent group[18], Dreamlet using only a fixed effects model shows statistical performance matching or exceeding the best performing current methods while giving the most accurate estimates of the simulated effect sizes (Fig. 2D, Supplementary Figs. 1–2). Using the initial weights from a Poisson model improves Dreamlet's performance compared to the unweighted version. We note that these simulations

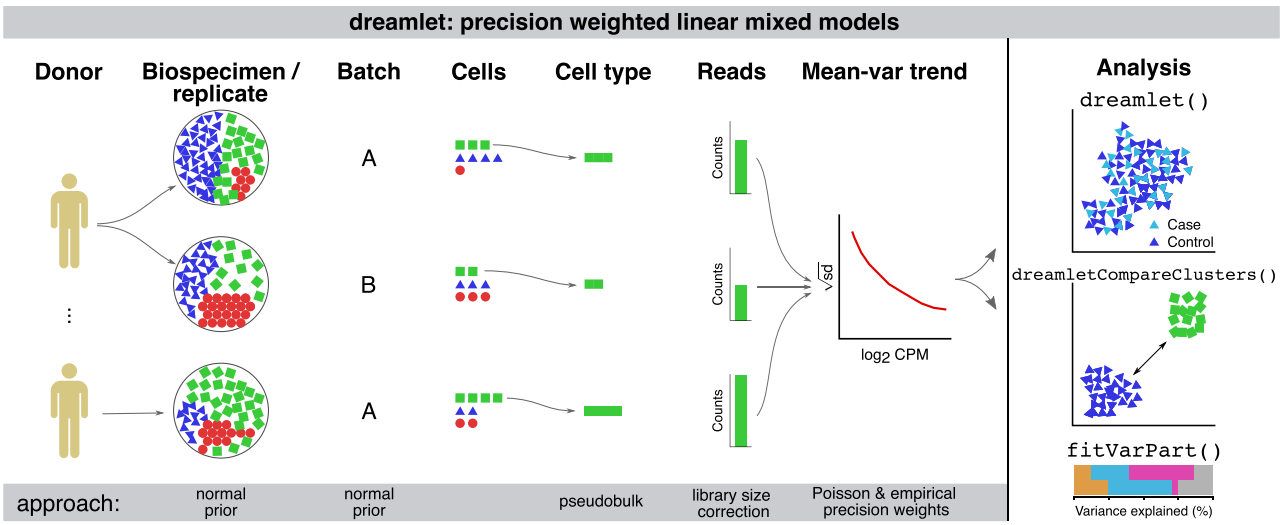


Fig. 1 | Illustration of dreamlet workflow using precision-weighted linear mixed models. Expression variation across multiple biological or technical replicates and technical batches is modeled using a random effect with a normal prior in a linear mixed model. Pseudobulk counts are computed for each cell type, and standard library size correction is performed. Precision weights are initialized using an approximation of a Poisson counting model and are then used in a second weighting step to model the empirical mean-variance trend. The dreamlet package provides interfaces for differential expression analysis across donors and cell clusters, for variance partitioning analysis, and for downstream analyses and visualization.

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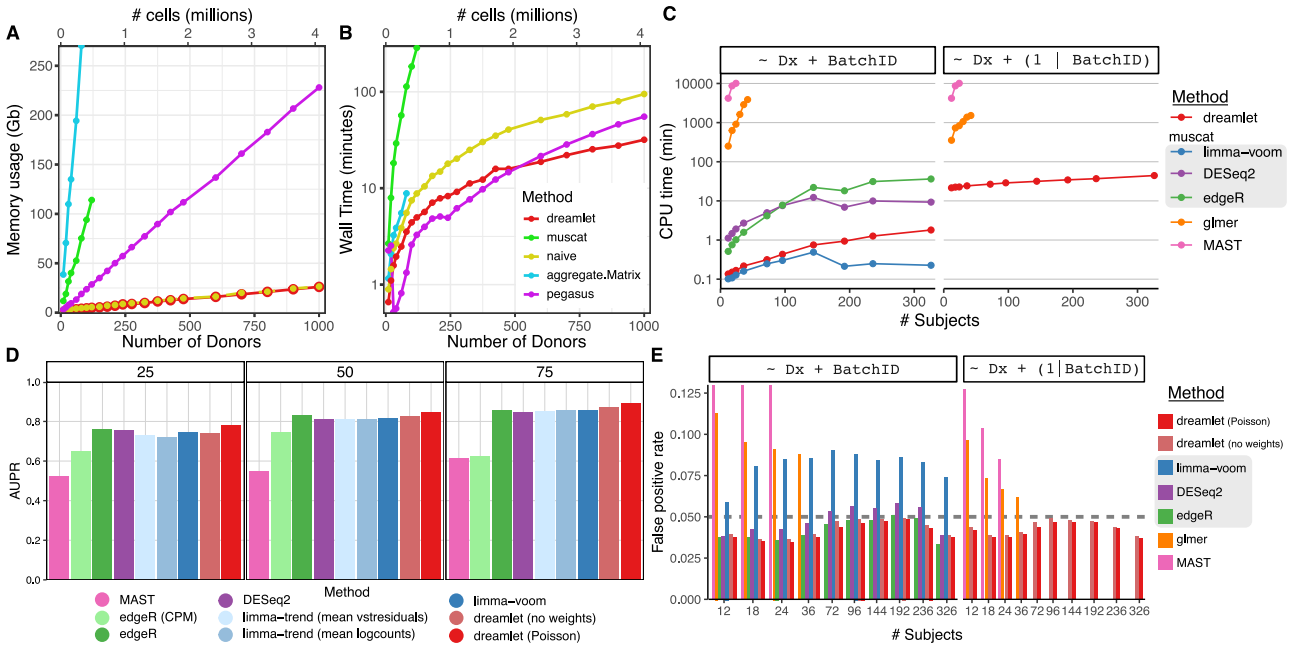


Fig. 2 | Computational and statistical performance of Dreamlet workflow. **A**, **B** Peak memory usage (**A**) and CPU time (**B**) averaged across ten runs using 1 CPU core on a machine with 512 Gb memory. **C** CPU times for differential expression analysis for an increasing number of subjects for six differential expression methods. Methods in the gray box use a pseudobulk approach run using muscat software, while GLMER and MAST model the data at the single cell level using generalized linear (mixed) models. **D** Performance of nine differential expression methods with sample size increasing from 25 to 75 subjects measured by area under the precision recall curve (AUPR). **E** False positive rates for an increasing number of subjects using real single-nucleus RNA-seq from postmortem human brains and permuting the disease status. Dashed horizontal gray lines indicate the target false positive rate of 5%. Results are shown for testing the effect of diagnosis, with batch modeled as a fixed effect (left) or a random effect (right). Methods in the gray box use a pseudobulk approach run with Muscat software, while GLMER and MAST model the data at the single-cell level. Source data are provided as a Source Data file.

used a simple study design in which all nine differential expression methods were applicable, whereas Dreamlet is the only scalable method capable of modeling complex study designs with random effects. Yet simulating single-cell transcriptomics data that accurately recapitulates the complexity of real datasets is notoriously challenging[18,29]. Instead, we used real data from human postmortem brains and permuted the disease status to estimate the empirical false positive rate across differential expression methods within each of 12 cell types. Of the methods that control the false positive rate across a range of sample sizes, Dreamlet is the only one that can model random effects (Fig. 2E). Results are similar when examined within each of the 12 cell types. Notably, the limma/voom method applied by muscat[18] shows inflated false positive rates in most conditions. Generalized linear mixed models used by the glmer[30] and MAST[21] methods can model random effects, but have an inflated false positive rate for small and moderate sample sizes. This is consistent with the fact that the null distributions of the coefficient estimates for these methods are normally distributed only in the asymptotic limit of a large sample size. Additional methods tested by others do not scale to these large datasets[18–20].

Multiplexing combines multiple samples within parts of the experimental workflow, reducing the cost of generating large-scale datasets. This process creates many small batches, each of 6–12 samples, that can share technical artifacts. Statistically, these batches are termed 'high-dimensional' when their number is large compared to the sample size. To control the false positive rate, a statistical model should account for these high-dimensional batch effects. Yet widely used fixed effects models can suffer from a substantial decrease in power, and can even perform worse when including the batch effect compared to omitting it from the model[31]. We show that using a linear mixed model to account for batches with a random effect controls the false positive rate while retaining power (Supplementary Fig 3).

Next, we apply Dreamlet to single-cell RNA-seq from COVID-19 patients and bone metastasis from prostate cancer. We then describe a novel single-nucleus RNA-seq dataset from an Alzheimer's disease cohort, and perform analysis using Dreamlet to examine cell-type-specific biology of the disease.

Dreamlet uncovers cell-type-specific response associated with COVID-19 severity

The COMBAT Consortium[32] collected blood from patients hospitalized with COVID-19 or sepsis, as well as from non-hospitalized COVID-19 patients and healthy controls. We performed dreamlet analysis on single-cell transcriptome data of 674K cells from 110 donors to identify transcriptional responses to SARS-CoV-2 infection associated with COVID-19 severity. The immune response associated with infection status is substantial and varies across cell types (Supplementary Fig 4), so we focus on monocyte subtypes and related cell types due to their key role in inflammatory response (Fig. 3A). Analysis of pseudobulk counts for classical monocytes shows a strong mean-variance trend in the $\log_2$ counts which dreamlet models using two-step precision weights (Fig. 3B). Variance partitioning analysis for each gene in classical monocytes estimates the fraction of expression variance attributable to variation across six disease states (i.e., four COVID-19 severity levels, sepsis, and healthy control), age and sex (Fig. 3C). Expression variation across disease states was the strongest source of variation with a median of 11.4% and 3250 of the 12,472 genes with sufficient expression explaining more than 25% of expression variance. Genes show a number of variance partitioning profiles with *CLU* showing variation across disease states, PRDX2 showing variation across age, and *XIST* varying across sex since it is on the X chromosome (Fig. 3D). Comparing gene expression in classical monocytes between patients with mild COVID-19 and healthy controls identified 1824 differentially expressed genes (Fig. 3E). Key response genes showed different expression patterns, with *CLU* expression increasing with

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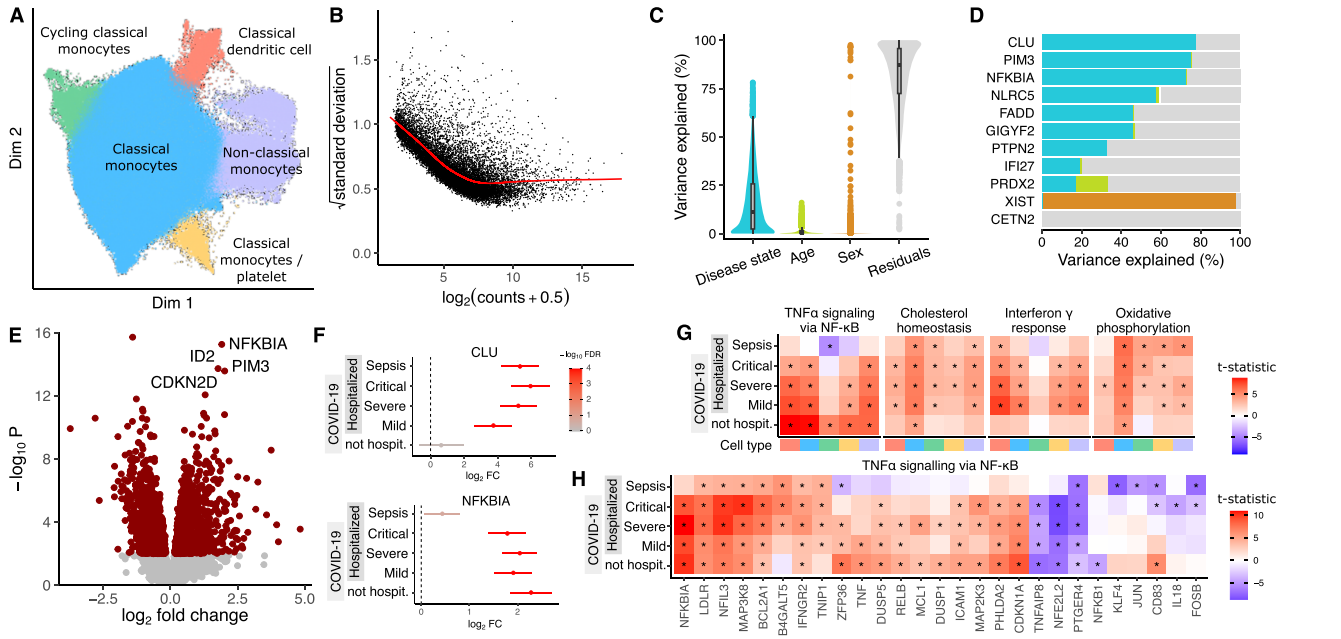


Fig. 3 | Dreamlet analysis of expression response associated with COVID-19 severity in monocyte populations. A Dimensionality reduction showing monocyte subtypes and related cell types. **B** Plot of mean-variance trend for expression counts in classical monocytes**. C** Violin plot summarizing variance partitioning analysis of classical monocytes separating the fraction of expression variation for each gene into four components. Box indicates first and third quartiles, whiskers indicate 1.5 inter-quartile range. **D** Representative genes with high variance fractions explained by each of the four components. **E** Volcano plot of differential analysis between mild COVID-19 and healthy controls. Red points indicate genes with FDR < 5% to account for multiple testing of the two-sided tests. **F** Forest plot showing $\log_2$ fold change of expression of *CLU* and *NFKBIA* in disease states compared to healthy controls. Color indicates false discovery rate after study-wide correction for multiple testing. Error bars indicate 95% confidence interval. **G** Heatmap of test statistics from gene set enrichment analysis shown for five disease states compared to healthy controls across five cell subtypes. Cell type color is indicated in (**A**). Study-wide FDR < 5% is indicated by '*'. **H** Heatmap of test statistics for a subset of genes in the gene set 'TNFα signaling via NF-κB' from MSigDB[74].

COVID-19 severity and sepsis compared to healthy controls while *NFKBIA* upregulated in COVID-19 but not sepsis patients (Fig. 3F). Gene set analysis using the full spectrum of test statistics[33] identified key immune response pathways activated in different cell types based on disease state (Fig. 3G). TNFα signaling via NF-κB showed activation in all COVID-19 disease states but not in patients with sepsis, while cholesterol homeostasis, interferon-γ and oxidative phosphorylation were activated in hospitalized COVID-19 patients but not those who were not admitted. Within the TNFα signaling pathway genes showed different patterns based on disease state, with *NFKBIA* upregulation specific to COVID-19 patients, *TNF* activation only in mild and non-hospitalized COVID-19 patients, and *CD83* upregulation specific to non-hospitalized COVID-19 patients (Fig. 3H).

Dreamlet identifies robust transcriptional changes in bone metastases from prostate cancer

Prostate cancer can metastasize to the bone and result in a very poor patient prognosis. Kfoury et al[11]. performed single cell RNA-seq on solid tumor, involved bone marrow, and distal bone marrow from nine prostate cancer patients with bone metastases, as well as benign bone marrow from seven patients without cancer. We applied the dreamlet workflow to identify genes that were differentially expressed based on disease status by modeling the multiple measurements from each cancer patient using a random effect. The number of expressed and differentially expressed genes varied widely across cell types and group comparisons (Supplementary Fig. 5). Analysis of pseudobulk counts for a monocyte cluster (Fig. 4A) shows the typical mean-variance trend that dreamlet models (Fig. 4B). Variance partitioning analysis in this cell type estimates the fraction of expression variance attributable to variation across patients, disease status of each sample (i.e., tumor, involved bone marrow, etc), and disease status of the patient (i.e., cancer vs non-cancer). Expression variation across patients is the strongest, while disease status explains greater than 10% of expression variation for 525 of 1899 (27.6%) genes with sufficient expression to be included in the analysis (Fig. 4C). This is consistent with cancer being a much stronger driver of gene expression changes than TB, above. Genes show a range of variance partitioning profiles (Fig. 4D). For example, while *CCNL1* shows high variance across disease status, *VIM* has high variation explained by patient status (Fig. 4E). Dreamlet analysis identified 157 genes in this monocyte cluster as differentially expressed between tumor and involved bone marrow at a study-wide FDR of 5% (Fig. 4F). Gene set analysis using the full spectrum of test statistics identified upregulation of MHC class II proteins and protein folding chaperones in the tumor samples across a range of cell types, including multiple monocyte clusters (Fig. 4G). Examining protein folding chaperones shows upregulation in tumor samples for most genes with sufficient expression (Fig. 4H).

Modeling multiple sources of expression variation in a large Alzheimer's Disease cohort

We generated single-nucleus RNA-seq (snRNA-seq) on tissue samples from the dorsolateral prefrontal cortex (DLPFC) of postmortem brains from donors in the Mount Sinai NIH NeuroBioBank (Fig. 5A). Samples were multiplexed by pooling six donors using the nuclei hashing method[34]. Each pool was processed in duplicate using the 10x Genomics single-cell gene expression platform to produce technical replicates. Following raw data processing and quality control, the dataset comprised 586 samples from 299 donors over 60 years of age and 1.4 M nuclei. Of these donors, 150 had Alzheimer's disease (AD), and 149 were age-matched neurotypical controls (Supplementary Fig. 6). Cell cluster annotation identified 22 cell types, including eight subtypes of excitatory neurons and 6 subtypes of inhibitory neurons

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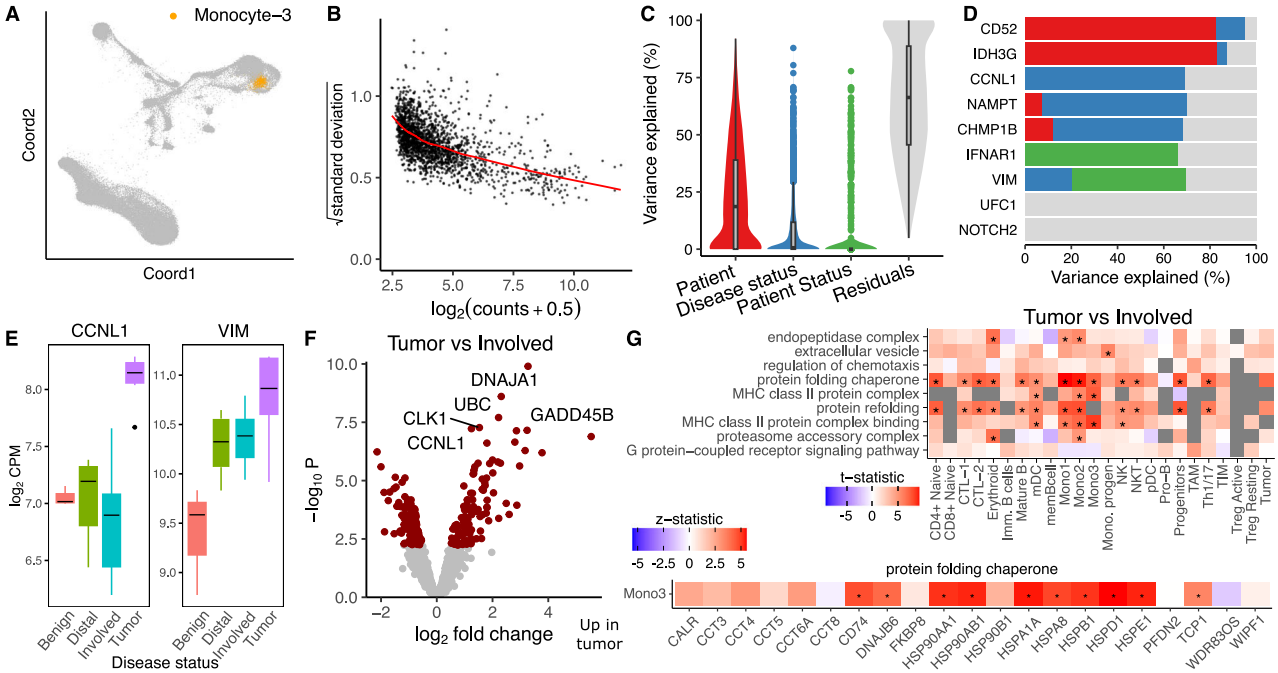


Fig. 4 | Dreamlet analysis of expression differences in prostate cancer bone metastases. A Dimensionality reduction highlighting monocyte population. **B** Plot of mean-variance trend for expression counts in this monocyte population. **C** Violin plot summarizing variance partitioning analysis separating the fraction of expression variation for each gene into four components. Box indicates first and third quartiles, whiskers indicate 1.5 inter-quartile range. **D** Representative genes with high variance fractions explained by each of the 4 components. **E** Representative genes with high expression variation across disease state (*CCNL1*) and subject cancer status (*VIM*). Box indicates first and third quartiles, whiskers indicate 1.5 inter-quartile range. **F** Volcano plot of differential analysis between tumor and involved bone marrow within each subject. Red points indicate genes with FDR < 5% to account for multiple testing of the two-sided tests. **G** Gene set analysis using the full spectrum of test statistics shows cell-type-specific signatures of tumor vs involved bone marrow. Study-wide FDR < 5% is indicated by '*'.

(Fig. 5B). Analysis of pseudobulk counts for microglia (Micro_PVM) shows the strong mean-variance trend (Fig. 5C) that dreamlet models with precision weights. Variance partitioning analysis for each gene in this cluster estimates the fraction of expression variance attributable to variation across technical replicates from the same subject, as well as variation across age, AD status, sample pool and sex (Fig. 5D). This identifies genes with different variance profiles where, for example, 20.2% of the variation in *PTPRG* is explained by Alzheimer's status while 97.3% of variance in NXPE1 is explained by sample pool (Fig. 5E).

In microglia, the median fraction of variation explained by subject is 38.4%, indicating good reproducibility in expression values across technical replicates. The effect of AD status is modest, with only 101 genes explaining more than 5% of the variance, underscoring the need for large sample sizes to characterize expression changes associated with the disease. While the sample pool explains minimal variance for most genes, 932 genes have >5% variance explained across the pools. This indicates a substantial batch effect for these genes. The high dimensionality of the batch effect due to multiplexing six samples per pool indicates that modeling this as a random effect using dreamlet can retain high power while avoiding false positive findings. These findings are consistent across other cell types (Supplementary Fig. 7). Further characterizing the batch effect reveals significant correlations between the variance across sample pools and the GC content of each gene (computed using all exons in the reference genome) in many cell clusters. This is consistent with previous work showing PCR artifacts driving technical variation[35,36] (Fig. 5F).

While single-cell/nucleus assays are uniquely capable of identifying cell-type-specific effects, the statistical power for each cell cluster in a given dataset varies widely. In particular, increasing the average number of nuclei per subject is directly related to an increase in the technical reproducibility of the gene expression measurements according to the variance explained across technical replicates from the same subject (Fig. 5G). Moreover, the number of nuclei per subject impacts the read count per subject, the number of genes that pass a minimum expression cutoff, and the number of genes that are found to be differentially expressed between AD and controls (Fig. 5H).

Large effect up-regulation of *PTPRG* in microglia of Alzheimer's disease cases

The number of expressed genes and differentially expressed genes between donors with AD and controls varies widely between cell types and increases with the number of nuclei observed per subject (Supplementary Fig. 8). The role of microglia in AD has received much recent attention due to findings from human genetic studies and mouse models[37–39], but the understanding of disease-associated gene expression changes has been more limited. Here, microglia have 1037 differentially expressed genes passing the study-wide 5% FDR cutoff (Fig. 6A). Even so, *PTPRG* stands out with a $\log_2$ fold change of 1.59 and *p*-value of 2.94e-28. Genes demonstrate a range of cell type specificity patterns with many differentially expressed genes shared across multiple excitatory and inhibitory neuron subtypes (Fig. 6B). Notably, the *PTPRG* up-regulation with this large effect size is specific to microglia, with significant but much smaller $\log_2$ fold changes observed in 1 excitatory and 2 inhibitory neuron subtypes (Fig. 6C). Other genes show effects across a range of cell types, as *PDE10A* is significantly down-regulated in AD in 13 of 14 neuron subtypes (Fig. 6D). Gene set analysis using the full spectrum of test statistics in each cell cluster identifies up-regulation of synapse assembly and glutamate pathways in subsets of excitatory and inhibitory neurons, up-regulation of presynapse organization and synaptic vesicle endocytosis most strongly in EN_L2_3_IT, neuronal action potential most strongly in IN_PVALB, IN_PVALB_CHC and IN_LAMP5 (Fig. 6E). In astrocytes there is a specific upregulation of protein polymerization. Oligodendrocyte precursors (OPC) have a specific upregulation of neural nucleus development.

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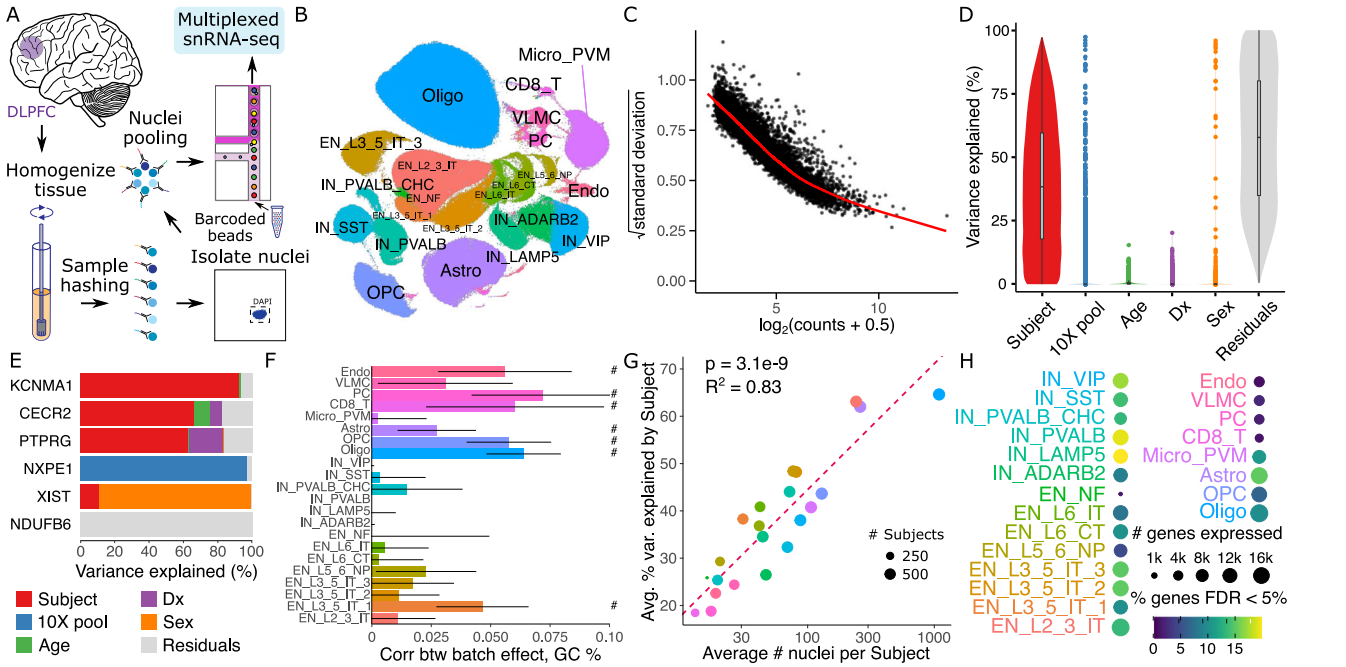


Fig. 5 | Dreamlet analysis of expression differences in Alzheimer's disease. **A** Multiplexed single-nucleus RNA-seq was performed on postmortem brain samples. **B** UMAP dimensionality reduction with cell cluster annotations. **C** Plot of mean-variance trend for expression counts in the microglia population. **D** Violin plot for microglia summarizing variance partitioning analysis separating the fraction of expression variation for each gene into six components. Box indicates first and third quartiles, whiskers indicate 1.5 inter-quartile range. **E** Representative genes with high variance fractions explained by each of the six components in microglia. **F** Spearman correlation of variance explained by sample pool ID with GC content of each gene. Bars indicate 95% confidence interval, '#' indicates <5% FDR. **G** Annotated clusters with more nuclei per subject show higher concordance in technical replicates. Circle size indicates the number of subjects with at least 10 nuclei observed for the cluster. *P*-value from linear regression is shown. **H** Number of genes passing expression cutoffs and the fraction of genes differentially expressed between AD subjects and controls at 5% FDR. Cell type abbreviations are: endothelial cells (Endo), vascular leptomeningeal cell (VLMC), pericyte (PC), CD8 + T-cells (CD8_T), microglia and perivascular macrophages (Micro_PVM), astrocytes (Astro), oligodendrocyte precursor cells (OCP), oligodendrocytes (Olig), inhibitory neurons (IN) and excitatory neurons (EN). IN annotations are followed by a marker gene, and EN are followed by a subtype annotation.

Notably, microglia have a specific up-regulation of the p38MAPK cascade, which is involved in microglial inflammatory response[40]. This pathway includes *DUSP10*, which is one of the top upregulated genes in microglia (Fig. 6F).

Discussion

The advent of single-cell technology has enabled the generation of large-scale high-resolution atlases to characterize cell-type-specific biology[41–43]. As the scale of datasets continues to increase, there is new potential to study variation across subjects at the cell-type level and how gene expression changes relate to a subject's age, sex, disease state, and many other traits[6–8,10].

We present the Dreamlet software, an open-source R package that enables analysis of massive-scale single-cell/nucleus transcriptome datasets. Dreamlet addresses both CPU and memory usage limitations by performing preprocessing and statistical analysis in parallel on multicore machines and by distributing work across multiple nodes in a compute cluster. Dreamlet also uses the H5AD format for on-disk data storage to enable data processing in smaller chunks to dramatically reduce memory usage[44]. The dreamlet workflow easily integrates into the Bioconductor ecosystem[45], and uses the SingleCellExperiment class[46] to facilitate compatibility with other analyses. Fitting precision-weighted linear mixed models[23] enables control of the false positive rate while retaining high power, even in the presence of high-dimensional batch effects. Beyond differential expression testing, Dreamlet provides seamless integration of downstream analysis, including quantifying sources of expression variation[47], gene set analysis using the full spectrum of gene-level t-statistics[33] and visualizing results.

We also introduce a novel dataset of 1.4 M single nuclei from postmortem brains from 150 Alzheimer's disease cases and 149 controls. Analysis using the Dreamlet software examines the cell-type-specific biology of AD. Highlighting the role of microglia in the disease, we observed that *PTPRG*, a protein tyrosine phosphatase receptor, is upregulated in AD and stands out substantially from all other genes in terms of effect size and *p*-value. PTPRG is an inflammatory marker, but its role in the molecular etiology of AD is unclear[48,49]. Interestingly, *PTPRG* is among the three genes (the other two being *APOE* and *DYPD*) that are reliably upregulated in two out of three previous human microglia transcriptome studies[50]. Recent genome-wide association studies of AD do not identify risk variants in the region of the gene, suggesting that *PTPRG* upregulation is reactive and may vary with the stage and progression of AD[50]. The AD expression signatures identified here show high concordance with recent single-nucleus data from Mathys, et al[51]. reanalyzed here with the dreamlet workflow, with overexpression of *PTPRG* in microglia being the top finding in both studies (Supplementary Figs. 9, 10).

Comparing differential expression signatures from our AD data using dreamlet, limma, and DESeq2, we observed high similarity genome-wide, but many genes showed different log fold changes across methods (Supplementary Figs. 11, 12). We evaluated the concordance in AD expression signatures across datasets for analyses using dreamlet, limma and DESeq2. The correlation between estimated log fold changes across datasets was highest for Dreamlet compared to the other methods (Supplementary Fig. 13).

We also compared batch correction modeling 10X pool as a random effect to a simpler approach of running Harmony[52] and including the top five components as covariates in a regression model. We observed that 10X pool explains substantial expression variance not explained by Harmony components (Supplementary Figs. 14, 15). In addition, genes whose differential expression test statistics change the

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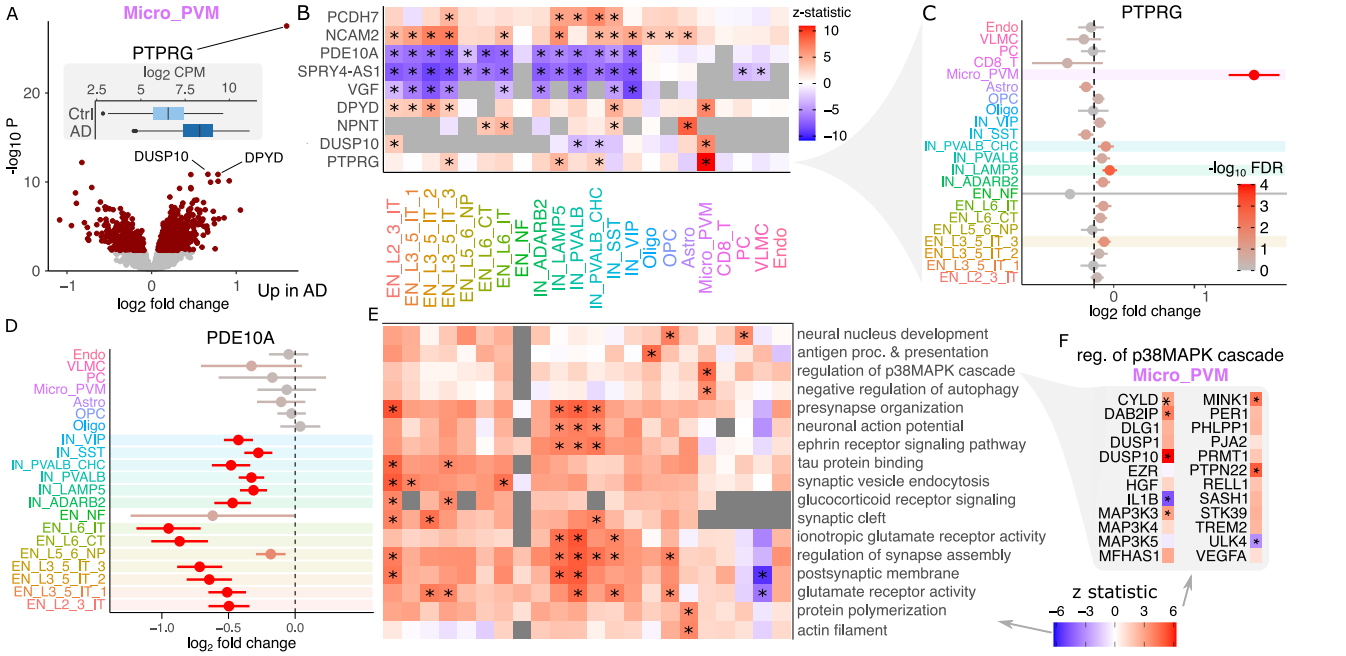


Fig. 6 | Gene expression signatures of Alzheimer's disease. A Volcano plot of differential expression between AD cases and controls in microglia. Red points indicate genes with FDR < 5% to account for multiple testing of the two-sided tests. The inset shows a box plot of PTPRG stratified by disease status. Box indicates first and third quartiles, whiskers indicate 1.5 inter-quartile range. **B** Heatmap showing differential expression z-statistic for genes in each cell cluster. '*' indicates study-wide FDR < 5% in all panels. Gray box indicates a gene did not pass the expression cutoff in that cell cluster. **C** Forest plot of $\log_2$ fold change for *PTPRG* in each cell type. Bars indicate 95% confidence interval. Color indicates FDR. **D** Forest plot for *PDE10A*. Bars indicate 95% confidence interval. **E** Gene set analysis using the full spectrum of differential expression test statistics. **F** Differential expression results for genes involved in the regulation of p38MAPK cascade.

most between analyses, including 10X pool versus Harmony components, have larger batch effects explained by variation across 10X pools (Supplementary Figs. 16–18). The findings underscore the importance of directly modeling batch-to-batch variation in gene expression in differential analysis using random effects.

Yet our work also highlights challenges in single-cell and -nucleus studies. First, technical batch effects can be substantial for some genes, and downstream analysis must account for them to control the false positive rate. Second, the findings within each cell cluster can be driven by biology as well as by limitations in the precision of gene expression measurements. We observe that increasing measurement precision by increasing the number of reads and nuclei improves reproducibility across technical replicates and increases the number of differentially expressed genes identified. This wide variation in measurement precision across cell clusters leads to substantial differences in the power to detect differential signals, even within the same dataset. A finding that a gene is differentially expressed in only one cell cluster may be driven by cell-type-specific disease biology, but could also reflect lower power in the other cell clusters. While the prospect of understanding the cell-type-specific biology of disease motivates these large cohort-scale studies, we recommend caution in the interpretation of cell-type-specific findings.

The dreamlet workflow focuses on differential expression across subjects by testing for changes in mean expression. Dreamlet, like all pseudobulk approaches, has some limitations. Aggregating gene expression across cells within a sample does not consider expression variation *within* a sample. Therefore, aggregating loses information about multimodal expression distributions and gene expression outliers within a sample, seen in cases of stimulus response[53,54], and continuous expression gradients seen in developmental trajectories and analyses of pseudotime and RNA velocity[17]. Addressing such challenges requires single-cell-level analyses.

In conclusion, we introduce dreamlet, an open-source R package (DiseaseNeurogenomics.github.io/dreamlet) that addresses previous challenges to perform efficient differential expression analyses across subjects in large-scale single-cell datasets. Dreamlet is substantially faster and uses less memory, supports complex statistical models and better controls the false positive rate compared to existing workflows, providing an important tool to address the need of expanding single cell/nucleus transcriptome datasets.

Methods

All research performed here complies with ethical regulations of the IRB at the Icahn School of Medicine at Mount Sinai and JJ Peters VA Medical Center.

Efficient computation of pseudobulk using on-disk memory

The dreamlet package creates pseudobulk data from the raw read counts for each sample and cell cluster stored in a SingleCellExperiment object[46]. SingleCellExperiment is Bioconductor's core data class for single-cell data, and data from other formats (i.e., Seurat[55]) can easily be converted to it. Dreamlet handles SingleCellExperiment objects to support storing large single-cell datasets, either in memory or on disk, in a way that is seamless for the end user.

For large-scale studies, loading the entire dataset into memory can be prohibitive. Storing 18 K genes across 2 M cells with double precision would require 288 Gb memory. If the dataset is so sparse that 80% of the entries are zero, loading the entire dataset as a `sparseMatrix` object requires 58 Gb of data. Yet, instead of loading the entire dataset into memory, the data can remain on disk and be accessed using an approach that takes advantage of the H5AD file format built on top of the HDF5 format[44]. The zellkonverter package[56] uses a DelayedArray backend to provide a seamless interface to an on-disk H5AD dataset through the interface of the SingleCellExperiment class. This enables any analysis designed for the SingleCellExperiment class to leverage on-disk access to large-scale datasets. This can dramatically reduce memory usage while still retaining high performance.

Creating pseudobulk from a large dataset involves summing reads across a set of cells for each gene. While on-disk access to H5AD avoids loading the entire dataset into memory at once, creating pseudobulk requires accessing each dataset entry. For large datasets, the amount of time required for this step can vary dramatically depending on implementation details. Following extensive experimentation using R, Rcpp and C++ code, we use the beachmat library[57] to summarize each block containing all genes and a subset of cells into pseudobulk. Using Dreamlet to compute pseudobulk from real single-cell data reduces both compute time and peak memory usage by more than an order of magnitude. For large datasets of hundreds of donors, only Dreamlet is tractable without purchasing an expensive high-memory machine.

Performance comparison for computing pseudobulk

H5AD files were created from real single-nucleus RNA-seq data generated here using nuclei annotated using eight cell-type clusters for 35k genes. Donors were sampled with replacement to evaluate performance on up to 1000 donors. We compared `aggregateToPseudoBulk()` in our dreamlet R package, `aggregate.Matrix()` in the Matrix.utils R package, `aggregateData()` in the muscat R package (which uses `scuttle::summarizeAssayByGroup()` in the backend), and `pegasus.pseudobulk()` in the pegasus python library[26].

We also implemented a 'naive' method that reads in single-cell data counts from the H5AD for one donor and cell type at a time. For eight cell types and 1000 donors, this reads data into memory in 8000 chunks to keep memory usage low. Once the data is in memory, pseudobulk is computed by summing across nuclei for each gene. Because H5AD data is stored in a distributed format, these values are distributed throughout the file. Accessing these chunks requires many inefficient read operations. We note that `aggregateToPseudoBulk()` consolidates these into fewer operations.

Each method was run ten times for each condition, and the average memory usage and CPU time are shown. Performance was assessed on a compute cluster where each run had access to 512 Gb memory. For a fair comparison, results are shown using only one thread to compute the pseudobulk values. (Dreamlet can use more threads to achieve faster performance at the expense of using more memory.)

We note that peak memory usage of native R code can vary dramatically based on available system memory and when R chooses to invoke garbage collection. Despite substantial fluctuations, especially for memory usage, these performance results are very robust across multiple runs and changes to internal parameters.

Precision-weighted linear mixed models

In widely cited work on differential expression analysis of RNA-seq data, Law, et al. [24] demonstrate the feasibility of modeling measurement uncertainty in a count response by weighting by the precision (i.e., reciprocal of observation-level variance). Importantly, they show that approximating the log-transformed counts using a weighted linear regression can outperform NB regression that models counts explicitly but suffers from poor hypothesis testing for finite sample sizes (since the null distribution of the test statistics relies on asymptotic theory). Our motivation for using weighted linear regression to model transformed counts follows that of Law, et al[24], but becomes more pressing with repeated measures and complex study designs. In previous work on bulk RNA-seq, we have demonstrated that precision-weighted linear mixed models are computationally efficient and hypothesis tests have good finite-sample performance in retaining power while controlling the false positive rate[23]. Using generalized linear mixed models (GLMM), such as a negative binomial mixed model, can be very computationally demanding, suffer from convergence issues in real data, and produce poorly calibrated p-values on finite samples.

While Law, et al.[24] consider precision weights to model heteroskedasticity of bulk RNA-seq counts, here we consider two levels of precision weights.

Modeling measurement uncertainty in count models using a two-stage weighting approach

The original biospecimen is often composed of thousands of cells of a particular type, or corresponding to a particular empirically defined cell cluster. To study the biology of a given cell type, experimental workflows randomly sample single cells for RNA sequencing. Since this is a stochastic process that samples cells from a much larger population, gene expression measurements aggregated across more cells more precisely represent expression in the full population. Consequently, the precision of gene expression measurements across samples is directly related to the number of cells sequenced per specimen. Statistically, gene expression measurements vary in their precision and are thus heteroskedastic.

The limited read count per gene in RNA-seq experiments has a similar effect on measurement uncertainty, with more reads giving a more precise measurement on the log scale. The widely used limma-voom approach[24] estimates precision weights by fitting a linear model for each gene. Voom then smooths the relationship between the $\log_2$ counts per million and the square-root residual standard deviation from the model fit for each gene. We have previously extended the voom approach to enable fitting of linear mixed models for estimating precision weights[23]. Here, we further extend this work to model heteroskedasticity using initial precision weights. The resulting estimated precision weights incorporate measurement uncertainty arising from stochastic sampling at both the cell and read levels.

Here, we initialize the precision weights used in an empirical mean-variance trend fit by approximating a Poisson generative model of the pseudobulk read counts. See Supplementary Methods for details.

Empirical Bayes shrinkage for precision-weighted linear mixed models

For small sample sizes, parameter estimates can have high sampling variance. In seminal work, Smyth[25] developed an empirical Bayes approach that borrows information across genes to estimate the residual variance. The widely used limma package[58] fits a linear model for each gene, performs the empirical Bayes step, and then computes a moderated t-statistic with a modified null distribution. In the case of a linear model, Smyth's empirical Bayes method uses a conjugate prior on the residual variances and assumes they are drawn from an inverse gamma distribution (i.e., precisions are drawn from a scaled chi-squared distribution) with parameters estimated from the data. A key value in this calculation is the residual degrees of freedom.

In the case of a linear model with n samples and p covariates (including the intercept), the residual degrees of freedom (df_r) is simply $n - p$. However, the case of a linear mixed model used here is more complicated. In this case, we show that the residual variance estimates follow a distribution given by a weighted mixture of n chi-squared random variables, where the weights depend on both the data and the estimated model parameters (Supplementary Information). We match the expected value of this mixture distribution using a single chi-square and use its degrees of freedom to approximate the df_r of the linear mixed model. Importantly, this method is exact for linear models, approximate for linear mixed models with any number of random effects, and the approximation improves with sample size.

Analysis of simulated single-cell data

We followed the workflow from Cromwell, et al.[18] to estimate parameters from read single cell data and then simulate data based on this. We integrated our Dreamlet software into their existing workflow and used the performance metrics described in their paper. We also added code to compute the area under the precision-recall curve (AUPR) and F1 scores to evaluate statistical performance. Finally, we extended the simulations to vary the number of cells observed per sample to better match real data. Instead of specifying the exact number of cells

observed per sample, the average number is specified, and the number for each sample is drawn from a Dirichlet-multinomial distribution.

Analysis of real single-nucleus data with permuted disease labels

We used the single-nucleus data generated here and randomly permuted the disease labels of each donor while retaining the same fraction of Alzheimer's disease cases and controls within each sample batch. Each method was then run with default parameters.

Single-nucleus RNA-seq data generation

Study cohort. Frozen brain tissue samples derived from DLPFC (Brodmann area 9/46) were obtained from the Mount Sinai Brain Bank (MSBB–Mount Sinai NIH Neurobiobank), which holds over 2000 human brains. Since we wanted to leverage samples from donors with either no discernible neuropathology or cognitive complaints (controls), or with only AD-associated neuropathology, we narrowed our initial selection of brain donors using a combination of neuropathological and clinical criteria inspired by previous work[59,60]. AD samples needed to be classified by (1) CERAD protocol[61] as "AD possible", "AD probable" or "AD definite", (2) Braak AD staging protocol[62] within the stages 3–6, and (3) clinical dementia rating[63] within the rate 0.5-5. Furthermore, our subset of AD donors cannot be diagnosed with Parkinson's disease or Diffuse Lewy body disease. Conversely, control samples needed to be classified by (1) CERAD protocol as "no AD" or "possible AD" and (2) Braak AD staging protocol within the stage 0-2. Additionally, control samples cannot be diagnosed with any other neurodegenerative, neurological or neuropathological diagnosis. Neuropathological assessments, cognitive, medical status and neurological status were performed according to established procedures[64]. All neuropsychological, diagnostic and autopsy protocols were approved by the Icahn School of Medicine at Mount Sinai and JJ Peters VA Medical Center Institutional Review Boards.

Isolation and fluorescence-activated nuclear sorting (FANS) of nuclei with hashing

All buffers were supplemented with RNAse inhibitors (Takara). Six samples were processed in parallel. 25 mg of frozen postmortem human brain tissue from each specimen was homogenized in cold lysis buffer (0.32 M Sucrose, 5 mM $CaCl_2$, 3 mM Magnesium acetate, 0.1 mM, EDTA, 10 mM Tris-HCl, pH8, 1 mM DTT, 0.1% Triton X-100) and filtered through a 40 μm cell strainer. The flow-through was underlaid with sucrose solution (1.8 M Sucrose, 3 mM Magnesium acetate, 1 mM DTT, 10 mM Tris-HCl, pH8) and centrifuged at 107,000 g for 1 hour at 4 °C. Pellets were resuspended in PBS supplemented with 0.5% bovine serum albumin (BSA). Resuspended nuclei were quantified (Countess II, Life Technologies) and 2 M from each sample were pelleted at 500 g for 5 min at 4 °C and re-suspended in 100 μl staining buffer (2% BSA, 0.02% Tween-20 in PBS). Each sample incubated with 1 μg of a distinct TotalSeq-A nuclear hashing antibody (Biolegend) for 30 min at 4 °C. Prior to FANS, volumes were brought up to 250 μl with PBS and 7-AAD (Invitrogen) added to facilitate detection of nuclei. 7-AAD positive nuclei were collected in tubes pre-coated with 5% BSA using a FACSAria flow cytometer (BD Biosciences).

snRNA-seq and library preparation

Following FANS, nuclei were washed twice in staining buffer before being re-suspended in 22 μl PBS and quantified. Nuclei concentrations were normalized, and equal amounts from each sample were pooled together. Two aliquots of 60,000 pooled nuclei (i.e., 10,000 each) were processed in parallel using 3′ v3.1 reagents (10x Genomics). At the cDNA amplification step (step 2.2), reactions were supplemented with a hash-tag oligo (HTO) cDNA "additive" primer (GTGACTGGAGTTCAGACGTGTGCTCTTCCGAT*C*T; *Phosphorothioate bond). Following cDNA amplification, supernatants from the 0.6x SPRI (Beckman Colter) selection step were retained for HTO library generation. Otherwise, cDNA libraries were prepared according to the manufacturer's instructions (10x Genomics). HTO libraries were prepared as described previously[65]. All libraries were sequenced at NYGC using the Novaseq platform (Illumina).

Single-nucleus data processing and quality control

Alignment and demultiplexing. Sequencing reads from all pools of multiplexed samples were aligned to the hg38 reference genome using STARsolo[66,67]. To assign the cells from each pool to their respective donors, we applied a genotype-based demultiplexing approach followed by genotype concordance. First, cellSNP[68] was used to pile up the expressed alleles from polymorphic sites overlapping snRNA-seq reads. Then, vireo[13] utilized those pile-ups to split cells into clusters corresponding to six distinct donors per pool. The identity of each cluster of cells to a particular donor was derived from genotype concordance analysis that compared the clusters of cells against reference SNP-array data using QTLtools-mbv[69]. While the majority of pools contained the cells from the expected sets of donors, we leveraged the genotype concordance results to detect and correct occasional sample swaps and mislabelings.

Quality control, UMAP, cell type annotation

QC. After genome alignment and demultiplexing, we applied rigorous three-step QC to remove ambient RNA and retain viable nuclei for downstream analysis. First, the QC was applied at the cell level. A battery of QC tests was performed to filter low-quality libraries and non-viable cells within each library. Poor-quality cells were detected by thresholding based on UMI counts, gene counts, and mitochondrial contents. We also checked for possible contamination from ambient RNA, a fraction of reads mapped to non-mRNA like rRNA, sRNA, and pseudogenes, as well as known confounding features such as lncRNA *MALAT1*. Further filtering was carried out by removing doublets using the Scrublet method[70]. Second, the QC is applied at the feature level. We removed features (genes) that are not robustly expressed by at least 0.05% of the cells/nuclei. Last, the QC was applied at the donor level. We remove donors with fewer than 50 cells, as they can introduce more noise into downstream analysis.

Batch correction. We have developed a tracking platform to record all technical covariates (such as 10x Genomics lot kit number, dates of different preparations, viable cell counts, etc.) and quality metrics derived from data preprocessing. We assessed the correlation between all pairs of technical variables using Canonical Correlation Analysis and used the Harmony method[52] to regress out the effect of sequencing pools before performing clustering and taxonomy analysis.

Clustering. Highly variable features were selected from mean and variance trends, and we used the k-Nearest-Neighbor (kNN) graph calculated on the basis of harmony-corrected PCA embedding space to cluster cells in the same cell-type using Leiden[71] clustering algorithm. We used UMAP[72] for the visualization of the resulting clusters.

Cellular taxonomy. Identified cell types will be annotated based on a combination of expert curation and machine-learning-based algorithms to query known gene marker signatures previously curated by Human Cell Atlas.

Sex check. For each donor, the labeled sex was checked to be consistent with expression on *XIST* and *UTY* genes on the X and Y chromosomes, respectively.

Statistics and reproducibility

Statistical analysis was performed with the Dreamlet software package in R. Reproducible analysis code is provided as described in Code Availability. Sample size was determined based on available data. Sample exclusion was performed using standard quality control metrics.

Data availability

The snRNA-seq data generated here is available at https://www.synapse.org/PsychAD_public with accession code syn51188606. Source data are provided with this paper.

Code availability

The dreamlet R package, including documentation, tutorials and code examples, is available at DiseaseNeuroGenomics.github.io/dreamlet and is available on Bioconductor at https://bioconductor.org/packages/dreamlet/. Code for simulations is available at github.com/GabrielHoffman/muscat-comparison_v2. Data analysis code and results for analyses in Figs. 3–6 are available at https://github.com/GabrielHoffman/dreamlet_analysis. Code has also been deposited to Zenodo and is available at https://doi.org/10.5281/zenodo.11453939[73] under Artistic-2.0 license. Software versions: dreamlet v1.1.24, muscat v1.11.2, DESeq2 v1.36.0, limma 3.52.1, edgeR 3.38.0, MAST 1.22.0, pegasus 1.10.0, STARsolo 2.7.9. cellSNP 1.2.0, vireo 0.5.6, QTLtools-mbv 1.3.

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Author contributions

G.E.H. developed the dreamlet package and performed analysis. V.H. provided tissue specimens from Alzheimer's disease brains and controls. A.H., C.C., M.A., Z.S. and S.A. generated novel snRNA-seq data from Alzheimer's disease brains and controls under the supervision of J.F.F. D.L., J.B., P.N.M., K.T., S.V. and G.V. performed processing of snRNA-seq data from Alzheimer's disease brains and controls generated here. G.E.H. and P.R. supervised analysis. G.E.H., D.L., J.B., J.F.F. and P.R. wrote the manuscript with input from all authors.

Competing interests

The authors declare no competing interests.

Additional information

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